Entity Relationship Schema & Data Model (ERD)

Decoupled Graph-Relational Architecture — PostgreSQL Implementation

Version: 1.0.0 **Date:** July 2026 **Status:** Approved Schema **Engine:** PostgreSQL 15+

1. Architecture Overview

The database architecture uses a **Decoupled Graph-Relational Model** in PostgreSQL. Personal metadata is stored independently in the `persons` table, while all direct genealogical connections are managed in a dedicated `relationships` junction table with status flags (`PENDING` vs `VERIFIED`).

2. Relational Schema DDL (PostgreSQL)

2.1 User Accounts Table (`users`)

```
CREATE TABLE users (
    id BIGSERIAL PRIMARY KEY,
    full_name VARCHAR(150) NOT NULL,
    email VARCHAR(255) UNIQUE NOT NULL,
    phone VARCHAR(50) UNIQUE,
    password_hash VARCHAR(255) NOT NULL,
    role VARCHAR(20) DEFAULT 'USER' CHECK (role IN ('USER', 'REVIEWER', 'ADMIN')),
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);
```

2.2 Family Individuals Table (`persons`)

```
CREATE TABLE persons (
    id BIGSERIAL PRIMARY KEY,
    first_name VARCHAR(100) NOT NULL,
    father_name VARCHAR(100),
    grand_father_name VARCHAR(100),
    family_name VARCHAR(100),
    gender VARCHAR(10) NOT NULL CHECK (gender IN ('MALE', 'FEMALE')),
    is_alive BOOLEAN DEFAULT TRUE NOT NULL,
    birth_date DATE,
    birth_year INT,
    death_date DATE,
    burial_place VARCHAR(255),
    photo_url TEXT,
    biography TEXT,
    created_by_user_id BIGINT REFERENCES users(id) ON DELETE SET NULL,
    claimed_by_user_id BIGINT UNIQUE REFERENCES users(id) ON DELETE SET NULL,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);
```

2.3 Direct Relationships Table (`relationships`)

```
CREATE TABLE relationships (
    id BIGSERIAL PRIMARY KEY,
    person_id BIGINT NOT NULL REFERENCES persons(id) ON DELETE CASCADE,
    related_person_id BIGINT NOT NULL REFERENCES persons(id) ON DELETE CASCADE,
    relationship_type VARCHAR(20) NOT NULL CHECK (relationship_type IN ('PARENT', 'SPOUSE',
'CHILD')),
    status VARCHAR(20) DEFAULT 'PENDING' CHECK (status IN ('PENDING', 'VERIFIED', 'REJECTED')),
    created_by_user_id BIGINT REFERENCES users(id) ON DELETE SET NULL,
    verified_by_user_id BIGINT REFERENCES users(id) ON DELETE SET NULL,
    verified_at TIMESTAMP WITH TIME ZONE,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT unique_relationship UNIQUE (person_id, related_person_id, relationship_type)
);
```

2.4 Branch Reviewers Assignment Table (`branch_reviewers`)

```
CREATE TABLE branch_reviewers (
    id BIGSERIAL PRIMARY KEY,
    user_id BIGINT NOT NULL REFERENCES users(id) ON DELETE CASCADE,
    root_person_id BIGINT NOT NULL REFERENCES persons(id) ON DELETE CASCADE,
    assigned_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT unique_reviewer_branch UNIQUE (user_id, root_person_id)
);
```

2.5 Deduplication & Merge Requests Table (`merge_requests`)

```
CREATE TABLE merge_requests (
    id BIGSERIAL PRIMARY KEY,
    primary_person_id BIGINT NOT NULL REFERENCES persons(id) ON DELETE CASCADE,
    duplicate_person_id BIGINT NOT NULL REFERENCES persons(id) ON DELETE CASCADE,
    status VARCHAR(20) DEFAULT 'PENDING' CHECK (status IN ('PENDING', 'APPROVED', 'REJECTED')),
    requested_by_user_id BIGINT REFERENCES users(id) ON DELETE SET NULL,
    reviewed_by_user_id BIGINT REFERENCES users(id) ON DELETE SET NULL,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);
```

3. Entity Relationships & Cardinality

- **Users to Persons (`users` $ ightarrow$ `persons`):** One-to-Many ($1 : N$) for node creation (`created_by_user_id`), and unique One-to-One ($1 : 1$) for profile claiming (`claimed_by_user_id`).
- **Persons to Relationships (`persons` $ ightarrow$ `relationships`):** Many-to-Many ($M : N$) self-referential association managed via the junction table storing direct relation type and status.
- **Reviewers to Branches (`users` $ ightarrow$ `branch_reviewers` $ ightarrow$ `persons`):** Links a steward user to a root ancestor node, granting verification authority over all descendant sub-tree nodes.